Final Report

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Data Science in Practice

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Table of Contents

1 Introduction 3

2 Project Objectives 4

2.1 Objectives & Motivations 4

2.2 Research Questions 4

3 Related Work 5

4 Data Considerations 6

4.1 Data Collection 6

4.2 Data Preparation 6

5 Results 7

5.1 RQ1: Title of RQ1 7

5.2 RQ2: Title of RQ2 7

5.3 RQn: Title of RQn 8

6 Conclusions 9

7 Latex Pointers 13

7.1 Figures 13

7.2 Code Listing 13

7.3 Math 14

7.4 References 14

Abstract

Provide a short description here (about 200-500 words) of your project. Do not go into detail, but offer a strategic overview of the project's aims. An abstract should whet the appetite for what comes next, not quench it with details.

Chapter 1: **Introduction**

Your first paragraph should set the stage for your project. What is it about? Why is it interesting? You should also briefly touch on what has been done on the topic before, but only very briefly as you will come back to this later in Chapter 3.

The next paragraph should briefly summarise what you hope to achieve in terms of the main research questions and the primary sources of data. Don't go into too much detail because you will do that in Chapter 2.

The final paragraph should briefly summarise the structure of the report in terms of the chapters that follow. In the next chapter we will ... Finally in Chapter 6 we will summarise what has been achieved and provide pointers for future work.

This report structure is based on the Final Year Project Template that Aonghus Lawlor has created, and that you will be using next year. I have adapted it for your needs. Working with LateX and Overleaf will feel quite alien and cumbersome at first but you will get used to it and in the long-run it is well worth the effort. You don't need to stick to this structure but something like this should serve you well. Note that the end of this file contains some helpful Latex pointers. Obviously you should remove these from the final version of your report.

Chapter 2: **Project Objectives**

This is where you get to tell us in detail about your project: why you chose it, who will be interested, why you think it is important, and what they main focus will be. If there are special challenges that you think are not immediately obvious then call attention to them in this chapter.

2.1 Objectives & Motivations

What motivated you to work on this particular project? Is it a passion project or an interest you have? By carrying out this project what do you hope to achieve?

2.2 Research Questions

Summarise the key research questions here. Try not to be vague. Try to present your RQs as hypothesese for example. You will come back to your RQs in detail in Chapter **??** so for now you really just want to provide the headline information. You can use a list such as the following for this:

1. RQ1 – Brief explanation of RQ1.
2. RQ2 – ... : :
3. RQn – Brief explanation of RQn

Chapter 3: **Related Work**

Your project does not exist in a vacuum, and its central problem, or a variant thereof, may have been tackled by others before you. In this section, you should describe how previous approaches have tackled the problem, and clearly articulate the state of the art (or SOA) for your project.

Make sure you learn how to correctly cite work such as [1] and notice how your bibliography gets built by LateX. You need to add papers/articles that you wish to cite in the references.bib file as special bibtex entries. Try using Google Scholar and click on the bibtex links to get this data; there are many other ways too such as using a bibliographic manager such as Mendeley (useful for final year but may be overkill for now).

Chapter 4: Data Considerations

In this section you should clearly introduce your datasets. Talk about why you chose these datasets, how you collected them, what's in them, and how you prepared them for your analysis.

4.1 Data Collection

4.2 Data Preparation

Chapter 5: **Results**

?? In this chapter you will explain each of your main research questions in a separate section, describe the data you used to answer the question, the methods used, and the results found. You will also discuss the results to demonstrate that you understand how they provide an answer to your RQ and the implications of this answer.

5.1 RQ1: Title of RQ1

Present your research question in sufficient detail for the reader to understand. You may have a number of variations of a RQ that you can include in this section or you may decide to split them into sparate RQ sections. That's your choice but it needs to make sense.

5.1.1 Data & Method

Explain in detail the data you used to answer this question. How did you use this data to answer the question?

5.1.2 Results

Present your key findings, usually as a set of graphs.

5.1.3 Discussion

Explain what you can conclude from these results as they related to your RQ?

5.2 RQ2: Title of RQ2

Present your research question in sufficient detail for the reader to understand.

5.2.1 Data & Method

5.2.2 Results

5.2.3 Discussion

: :

5.3 RQn: Title of RQn

Present your research question in sufficient detail for the reader to understand.

5.3.1 Data & Method

5.3.2 Results

5.3.3 Discussion

Chapter 6: **Conclusions**

In this section you will sum up your report, draw some conclusions about your work so far, and make some general observations about the work to come. You may also use this opportunity to express points of view, or make factual claims, that are more pertinent here than in other sections of the report. If your project raises some ethical concerns, for example about how data or users are treated, then address them here in a thoughtful manner.

Acknowledgements

In your Acknowledgements section, give credit to all the people who helped you in your project.

Bibliography

1. Knuth, D. E. *Art of computer programming, volume 2: Seminumerical algorithms* (Addison-Wesley Professional, 2014).

Appendices

You can use your appendix to include information or data that you feel is not needed for your main report but that is still relevant to your project. This might include secondary results or additional results that are similar to those presented in the main report.

Appendix A – Individual Contributions

For group projects use this section to briefly summarise the contribution of each group member. Don't go overboard here but clearly indicate who was responsible for what RQ for example.

Appendix B – Datasets & Notebooks

Please list the filenames of the main datasets used in your project along with a brief (one-sentence) description of each. Also list the notebooks with a similarly brief summary of their purpose. You can indicate too who was responsible for which dataset/notebook by indicating the initials of the student beside the dataset or notebook file.

Datasets Used

1. dataset1.csv (BS) – the main dataset for RQ1 and RQ2 which includes accident rates by region and year.
2. ...

Notebooks Used

1. data_cleaning.ipynb (BS) – the main data cleaning notebook responsible for downloading the datasets from Kaggle, verifying their contents, and preparing them for processing.
2. ...

Appendix N – Supplementary Results

...

Chapter 7: **Latex Pointers**

This chapter contains some examples on the usage of latex. Do not include in your final report.

7.1 Figures

From time to time, it's necessary to add pictures to your documents. Using LaTeX all pictures will be indexed automatically and tagged with successive numbers when using the figure environment and the graphicx package. We can reference the figure below using its label like this: Fig. 7.1.

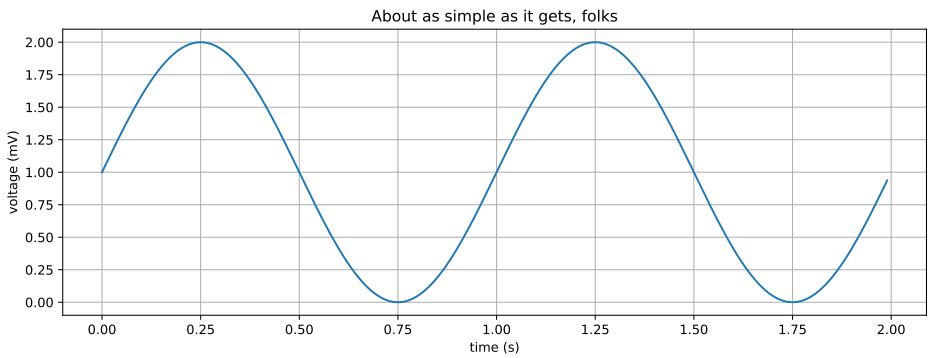

Figure 7.1: A sample graph

Lots more information in this tutorial.

7.2 Code Listing

You can list code using the *listings* package:

Listing 7.1: Borg Pattern

```
class Borg(object):
    __shared_state = {}

    def __init__(self):
        self.__dict__ = self.__shared_state
        self.state = 'Init'

    def __str__(self):
        return self.state
```

Lots more examples here.

7.3 Math

Here 7.1 is an example of including an equation

$$\lim_{x \to \infty} f(x) \tag{7.1}$$

More examples here and here.

7.4 References

Look up the bibtex references on google scholar or import from Mendeley or other reference managers. Add the bibtex snippet to the *references.bib* file. Then cite the reference like this:

As explained in [1], we also find that...

Lots more details here.